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MNC-Net: Multi-task graph structure learning based on node clustering for early Parkinson's disease diagnosis

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ABSTRACT

Purpose: The identification of early-stage Parkinson's disease (PD) is important for the effective management of patients, affecting their treatment and prognosis. Recently, structural brain networks (SBNs) have been used to diagnose PD. However, how to mine abnormal patterns from high-dimensional SBNs has been a challenge due to the complex topology of the brain. Meanwhile, the existing prediction mechanisms of deep learning models are often complicated, and it is difficult to extract effective interpretations. In addition, most works only focus on the classification of imaging and ignore clinical scores in practical applications, which limits the ability of the model. Inspired by the regional modularity of SBNs, we adopted graph learning from the perspective of node clustering to construct an interpretable framework for PD classification.

Methods: In this study, a multi-task graph structure learning framework based on node clustering (MNC-Net) is proposed for the early diagnosis of PD. Specifically, we modeled complex SBNs into modular graphs that facilitated the representation learning of abnormal patterns. Traditional graph neural networks are optimized through graph structure learning based on node clustering, which identifies potentially abnormal brain regions and reduces the impact of irrelevant noise. Furthermore, we employed a regression task to link clinical scores to disease classification, and incorporated latent domain information into model training through multi-task learning.

Results: We validated the proposed approach on the Parkinsons Progression Markers Initiative dataset. Experimental results showed that our MNC-Net effectively separated the early-stage PD from healthy controls(HC) with an accuracy of 95.5%. The t-SNE figures have showed that our graph structure learning method can capture more efficient and discriminatory features. Furthermore, node clustering parameters were used as important weights to extract salient task-related brain regions(ROIs). These ROIs are involved in the development of mood disorders, tremors, imbalances and other symptoms, highlighting the importance of memory, language and mild motor function in early PD. In addition, statistical results from clinical scores confirmed that our model could capture abnormal connectivity that was significantly different between PD and HC. These results are consistent with previous studies, demonstrating the interpretability of our methods.

1. Introduction

Parkinson's disease (PD) is a prevalent neurodegenerative disease. PD mainly changes the neurons in the substantia nigra region of the brain responsible for dopamine production, and affects the transmission of control signals for the body's movements [1]. Therefore, the clinical diagnosis of PD is mainly based on movement disorders, such as bradykinesia, resting tremors and imbalance, which are usually detected in the advanced stage and seriously affect the patient's life. In addition, non-motor symptoms such as depression, anosmia and cognitive decline often precede motor symptoms, but are often atypical, rendering early PD undiagnosed. However, early detection of PD and timely neuroprotective treatment may slow down disease progression and reduce the possibility of serious clinical symptoms [2]. Recently, machine learning (ML) algorithms have been widely used in medical diagnosis and have become an important tool for the analysis of neurological disorders. Therefore, it is very meaningful and feasible to construct a classification framework for the early diagnosis of PD using ML algorithms.

Diffusion tensor imaging (DTI) is an important tool in the study of neurological disorders. It is the most classical and widely used dMRI technology, which is sensitive to the abnormal microstructure of white matter [3]. A previous survey [4] found a correlation between DTI changes in multiple white matter regions and the conversion of motor symptoms, especially dopamine regions. This suggests the feasibility

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of longitudinal DTI as a potential biomarker to detect PD progression. Structural brain networks (SBNs) derived from DTI data have been used to analyze the imaging of neurological disorders, including PD [5]. Exploring the difference of SBNs between PD and healthy controls (HC) promotes the understanding of potential early lesions and improves the treatment methods. Therefore, our study primarily focused on the PD classification of DTI-based SBNs. Here, SBNs modeled the brain network as a graph that describes the connections or interactions (i.e., nodes and edges) between different discrete objects. Nodes are the Region of interests(ROIs) divided by the anatomy, and edges are the connections between them. Currently, this graph modeling approach has been commonly used for brain network representation in neurological disease diagnosis. However, there are two challenges to the brain network classification for PD diagnosis.

Challenge 1: High-dimensional noisy brain networks.

Whole-brain SBNs constructed from images tend to have high-dimensional connections and noise issues. Here, the noise is relative to the abnormal connection. Our goal was to investigate differences in structural connectivity between PD and HC. In terms of whole brain connections, the number of normal connections in PD is much greater than the number of abnormal connections. Due to individual differences, normal connections tend to differ in both PD and HC samples. This general difference may submerge abnormal connectivity and bring challenges to graph representation learning. The high-dimensional connections are derived from the complex topology of the brain itself and require a large number of nodes to represent. It may compromise graph distinguishability, leading to an overfitting problem.

Challenge 2: Limited model interpretability.

To a large extent, clinical decision-making requires the support of evidence collection and interpretation. The lack of evidence and interpretation makes the ML methods prediction results unacceptable to clinicians, particularly in disease diagnosis. In addition, interpretability may also explore the relationship between data and provide unique insights into the corresponding fields. However, due to the complex prediction mechanism of existing deep learning methods(DL), it is challenging to extract effective clinical interpretation from them [6,7].

To address the above issues, we proposed a multi-task node cluster based graph structure learning framework (MNC-Net) which combines graph structure learning and multi task representation learning to improve classification performance and identify potential brain regions.

We defined the disease classification as a graph representation task. Inspired by the modularity of the SBNs, we adopted a node clustering method for graph structure learning. Graph structure learning aggregates graph representation in a modular way, which facilitates the identification of abnormal network patterns and reduces the impact of individual difference noise.

For the advantage of graph neural networks (GNNs) which has the built-in interpretation, we employed GNNs as the backbone of the classification framework. Based on the clustering parameters of nodes, we could obtain relevant clinical interpretations by highlighting key nodes that are very favorable for prediction. In addition, unlike most GNNs methods that only focus on imaging data analysis and ignore the massive presence of scoring data in clinical diagnosis, we added clinical scores to graph representation learning in a multi-task manner. Specifically, our framework jointly learned the disease classification of SBNs and the regression of clinical scores. In this way, we introduced the correlation between clinical scores, brain networks and disease states into parameter training to improve the model's generalization performance through domain knowledge supervision.

Our contributions are as follows:

1. We proposed a multi-task graph representation learning framework for brain network classification in early PD. The algorithm combinesd the modularity of SBNs to aggregate efficient graph embeddings from high-dimensional noisy brain networks in a node-clustering manner.
2. In addition, we incorporated clinical scores information in a multi-task manner to guide feature learning under the latent space.
3. Furthermore, we provided significant brain regions and correlation analysis as model interpretability through parametric decomposition.

The remainder of this paper is organized as follows. Section 2 introduce the related studies of our work. In Section 3, we introduce the details of the proposed framework step by step. In Section 4, we introduce the dataset and experimental setup used in this paper, and then present comparative results. Section 5 investigate the choice of parameters for several key components of the proposed framework and discuss related interpretability. We conclude this work in Section 6.

2. Related work

DTI is a common non-invasive tool for studying PD. We can observe the structural integrity of the white matter through metrics such as fractional anisotropy (FA) which were derived from DTI. Furthermore, through fiber tract tracing, we obtained the structural connectivity (SC) between different ROIs, which construct SBNs.

2.1. Traditional machine learning methods(TML)

Early research often used TML methods such as support vector machine (SVM) and random forest (RF) to analyze DTI. Haller et al. [8] combined with TBSS and SVM to conduct inter group analysis and pattern recognition, and the accuracy rate of correctly classifying PD patients was as high as 97%. Prasuhn et al. [9] used bSVM to classify PD and normal people according to specific ROIs attributes. Yang et al. [7] proposed an ensemble learning framework that integrated the advantages of SVM, RF and other TML, and combined multimodal images and clinical scores to achieve accurate classification of PD (accuracy rate of up to 96.88%). However, these methods often rely heavily on the quality of artificial features, as well as contain complex feature selection, which is not conducive to the analysis of complex medical images.

2.2. Deep learning approach

With the development of deep learning(DL) in medical images, many methods have shown outstanding capabilities in disease diagnosis and analysis of neuroimaging, including convolutional neural networks (CNNs) and GNNs [10]. However, due to the locality of the convolution operator of CNN, the network can only learn the data structure of euclidean space, and is not suitable for irregular graph data. Its voxel based interpretations are difficult to conceptualize with entities and are not easy for clinical experts to understand [11]. GNNs has been a hot topic of deep learning in the past two years [12]. It is suitable for processing graph data and fusing irregular relationship information between entities (such as disease status, clinical score, etc.). In addition, GNNs is also widely used in the analysis of neural diseases due to its built-in explanatory characteristics.

Considering the complex topology of the brain and the noise, how to excavate the connection pattern of the diseased structure from the high-dimensional SBNs is not a trivial matter. Some works has attempted to solve this challenge by leveraging multi-perspective learning. Zhang et al. [13] learned perspective-invariant brain network patterns through multi-view SBNs constructed by different fiber tract tracing algorithms. Yao et al. [14] used multiple atlases to build SBNs, and extracts the complementary information of ROIs connections at different scales through a mutual learning mechanism. Although these methods have showed high accuracy on graph classification tasks. However, there is still a lack of further analysis of the source of high performance, namely model interpretability of linked entities (e.g., brain regions,

white matter fibers). Therefore, we proposed a graph structure learning GNNs based on node clustering, which can capture abnormal patterns in complex brain networks for disease classification. At the same time, the community based aggregation of node features can effectively preserve the graph structure and facilitate the extraction of GNN's built-in interpretations .

2.3. Explainability in GNNs

Explainability is an essential basis for disease diagnosis. Although the DL solution has achieved significant classification performance, its lack of interpretability makes it difficult to be accepted by the medical community. The general method of generating interpretation is to find the edge mask of a graph to highlight the important salient regions learned by the model [15,16]. Based on this, cui et al. [17] proposed an interpretable brain network classification framework and added a brain network explanation generator to highlight abnormal disease patterns learned by the model. Another popular approach is parameter decomposition, which measures the importance of input features by directly studying the model parameters. Li et al. [18] designed models based on the systematic partitioning properties of functional brain networks(FBNs), and extracted individual or group-specific biomarkers with a built-in interpretation of parameters. The above methods have performed well in previous works. However, the following problems are still present when they are extended to SBNs analysis: Generation-based methods often obtain explanatory graphs by adding edge perturbations, which ignores the importance of node features. It is also challenging to adapt to brain graphs with a large number of nodes. In contrast, the method based on parameter decomposition is more stable and can catch more common abnormal network patterns. The modularity of SBNs has been found in previous studies, but there is no clear division study at present. Therefore, we used unsupervised clustering to learn the community of SBNs, and mapped the cluster centrality to regional significance.

2.4. Multi-task learning

Recently, most DL methods have used clinical scores as integrated image features for a single task; thus, ignoring the potential relationship between data limits the model's performance. Multi task learning (MST) is an effective solution. It makes the main task generalization better by sharing a certain degree of parameters between different tasks. For example, Yu et al. [19] used MTL to improve the transfer learning efficiency of MRI models on CT data and achieved more accurate multi-type heart index evaluation; Hou et al. [20] improved the performance of segmentation tasks through the localization task of breast tumors. There are two main ways to implement MTL. The most common is the hard-constrained parameter sharing, where all tasks share a model, and only the different output layers are separated in the last feature layer. This approach effectively reduces the risk of overfitting and is simple to implement. Differently, each task in the soft constraint parameter method shares a unique model and parameter, and regularization promotes model learning through the similarity between tasks [21]. Considering the need of model interpretability, we choose the hard constraint sharing method to improve the generalization and reduce the complexity of the model.

3. Methods

3.1. Architecture overview

The overall processes of our proposed method are shown in Fig. 1. We defined PD diagnosis as a classification problem and proposed a framework for learning graph structures based on SBNs. Specifically, we formulated a unified three-stage brain network classification pipeline. It involves graph construction, graph classification, and interpretation extraction.

3.2. Preliminary work

3.2.1. Data processing

In the experiments, we used two kinds of data: MRI images and clinical scores, which required different preprocessing. First, the MRI images include DTI and T1, both of which were processed using PANDA toolboxes [22]. The specific pretreatment processes are as follows.

(1) Foreground extraction: Convert the image from DICOM to NIFTI, perform skull removal and crop the gap operation on the DTI and T1 to reduce irrelevant information. In addition, we performed eddy current correction for DTI to reduce the impact of artifacts in image acquisition. The T1 result will be registered to the standard MNI space.

(2) Tensor fitting: Tensor fitting of DTI results is obtained to get the FA's NIFTI file. Then, the FA will be registered to the standard MNI space [23] for subsequent network construction.

(3) Network node construction: Obtain an AAL template on the individual's DTI space, dividing the brain into 90 regions as the definition of the network nodes.

(4) Construction of network edges: Deterministic tracking algorithm is used to track fiber bundles to reconstruct the structural connections between various brain regions. The trace is stopped when the value of the FA between two voxels is less than 0.2 or the tracking angle is greater than 45°.

With the above steps, we obtained a 90×90 SBNs for each subject, which node is defined as the brain ROI, and the edge is the number of fiber bundle connections between the two ROIs. To reduce the effects of image noise during the tracking process, we normalized them.

Clinical scores is the doctor's assessment of a patient's clinical status through professional neurological scales. Through these scales, doctors make a very objective judgment on the patient's movement, spirit, degree of course development, and treatment effect. The specific assessment method for each symptom results in a different total score. Therefore, we divided them by their respective total scores as the target of the regression task.

3.2.2. Brain network modeling

In this study, the brain graph is modeled by the topological relationship between ROIs, which is used for mapping the SBNs to a learnable representation.

Suppose that the undirected brain graph we constructed is defined as $G = \{V, A, X\}$. $V = \{v_1, v_2, \ldots, v_N\}$ is a collection of nodes, and each node represents a brain region. N is the node quantity. A is the adjacency matrix defined as follows:

$$A_{ij} = \begin{cases} 0, & \text{if } v_j \notin V_i \\ 1, & \text{if } v_j \in V_i \end{cases} \tag{1}$$

where V_i is the neighbor set of node v_i, which are the top 10 nearest neighbors connected to node v_i. X is the node features, each raw represents the connectivity between node v_i and the remaining nodes. In the experiment, the node's connectivity to itself is set to zero.

3.2.3. Graph representation learning

To fully consider the features and topological information of graphs, we used GNNs for representation learning. GNNs can efficiently handle graph data on non-euclidean spaces. The core of GNNs is the messaging mechanism, which updates the central nodes representation by aggregating the neighbor node features to pass the surrounding information. Specifically, graph G_i is represented by the l-layer network embedding as:

$$G_i^{(l)} = \sigma(W^{(l)} G_i^{(l-1)} A_i^{(l)}) \tag{2}$$

where $W^{(l)}$ is the l-layer weight matrix, $A_i^{(l)}$ is the adjacency matrix of g_i in l-layer, and $\sigma()$ is a nonlinear activation function. Thus, the two-layer GNNs can be calculated as:

$$G_i^{(1)} = \sigma_1 W^{(1)}((\sigma_0 W_i^{(0)} X^{(0)} A_i^{(0)}) A_i^{(1)}) \tag{3}$$

Our approach takes this general framework and refines the process within it.

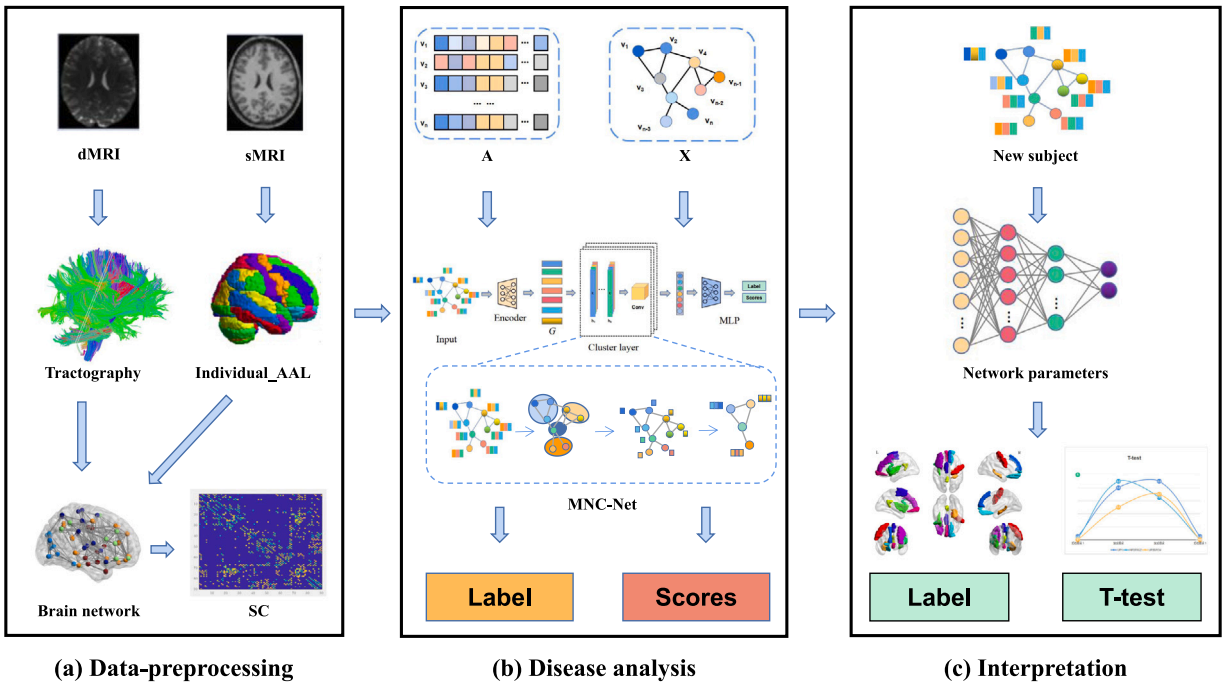


Fig. 1. The overview of our pipeline.(a) There is data processing of our proposed method; (b) we use MNC-Net to diagnose diseases in brain graph. A is the adjacency matrix, and X is the initial node feature matrix; (c) We extract disease-related salient ROIs as well as connectivity from the trained network layers.

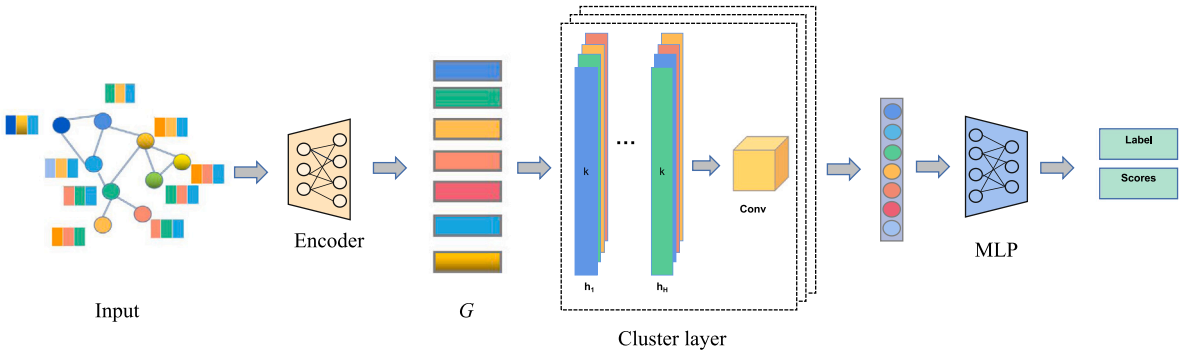


Fig. 2. The architecture of our methods. MNC-Net is composed of blocks of graph encoder and cluster-layer. It takes graph as input for graph structure learning. The graph encoder fuses the topology and attribute information of nodes into a initial graph embedding G, so that subsequent graph representation updates can be completed in one stage. The clustering layer updates G through structure learning. Specifically, the memory key k_i is used as the K clustering center, and the cluster probability of each node is calculated by the t distribution. The result of h-group clustering is then passed through a convolutional layer of 1*1 to obtain the unified representation and passed to the next layer. The output learned by stacking the cluster layers is passed to the readout function and MLP layer to obtain the predicted graph label or clinical scores.

3.3. Framework

As shown in Fig. 2, the proposed framework is composed of three components: the graph represents the encoder (G), the node cluster layer, and the 1*1 convolutional layer (conv).

Traditional GNNs have a large amount of computation. With the deepening of the network layer, it is easy to have over-smoothing problems, which limits the ability of the model. To improve this point, we adopted a node clustering module to achieve one-stage representation update. Specifically, the topology and attribute information of the graph are fused before inputting the model. The pooling and message transmission process can be replaced by the cluster layer that performs graph representation learning and node clustering at the same time. In this way, we highlighted salient connections in a clustering manner and reduced the impact of spurious connections.

3.3.1. Structure learning

We considered a brain graph as a set of node representations in constant order. Graph Encoder G fuses the attributes and topological information of the nodes into a graph embedding, which is eventually converted into a set of features in node units of g.

The input of G is the adjacency matrix A of graph and the node feature X, which is fused in the following way, and the initial representation of the graph g is $G^{(0)}$:

$$G^{(0)} = \sigma(Linear(A^{(0)})) \parallel \sigma(W_2\sigma(W_1X^{(0)})) \tag{4}$$

where W_1, W_2 are a learnable parameters for fusing the properties of nodes with topology information, and the initial weights in the experiment are set as the identity matrix. $Linear(\cdot)$ is a linear transformation function. $G^{(0)} = \{g_1, g_2, \ldots, g_N\}$ is a collection of node represents g_i. With graph fusion, pooling can be performed in conjunction with embedding updation. This greatly improves the learning efficiency learning and reduces the impact of message passing sequence changes on high-level graph representation.

SBNs and FBNs can be highly modular, and the modules are connected in a similar way internally and functionally [24]. The fmri studies have found that brain regions do not exist independently in FBNs, but interact with other regions, which is similar to community

aggregation [23]. Therefore, we speculated that the structural connections between brain regions also have similar characteristics to some extent. That is, there may also be such community characteristics in SBNs. Based on this assumption, we used node clustering to explore SBNs representations.

The clustering layer (C) is an unsupervised K clustering module that catch the node clustering patterns learned through training. This mode is stored as a memory key for updating graph representations with node features. The specific process is as follows: First, using the stored memory keys to obtain a set of centroids $(k_1, k_2 \dots k_K)$ according to the current graph, K is the number of keys, and we discussed its selection in Section 4.2.2. Second, the t-distribution is used to calculate the clustering probability of each node by the centroid as the score for node importance. The cluster probability Q_{ij} of v_i clustering into cluster k_j is as follows:

$$Q_{ij} = \frac{(1 + \|g_i - k_j\|^2/\tau)^{-\frac{\tau+1}{2}}}{\sum_{j'=0}^{K}(1 + \|g_i - k_{j'}\|^2/\tau)^{-\frac{\tau+1}{2}}} \tag{5}$$

where τ is the degree of freedom of the t-distribution, and $\|.\|^2$ is L2 regularization. Considering that, ROIs have different connection modes when performing different functions in FBNs, we extended a group of cluster center keys to multi-head clusters H, to further improve the learning ability of model. Thus, we obtained a set of attention tensors $(Q_{(1)} \dots Q_{(H)})$. To simplify network complexity, a $[1 * 1]$ convolutional layer ($conv$) was used to aggregate multi-head attention. Eventually, an attention matrix based on node clustering is as follows:

$$Q^{(l)} = softmax\left(\left(Q_{(1)} \| Q_{(2)} \dots \| Q_{(H)}\right)\right) \tag{6}$$

where $\|$ is the concatenation operation.

Finally, we mapped the input graph to a coarser representation space via the attention matrix:

$$G^{(l+1)} = \sigma(Q^{(l)\top} G^{(l)} W) \tag{7}$$

where $G^{(l)}$ is the input to the l-layer network, and W is the network parameter. To this, the clustering layer completed a graphical representation of the update.

3.3.2. Multi-task prediction

The output of the model includes the results of two tasks: the classification of individual brain graph and the score regression.

For the graph classification task, we obtained the prediction labels through the multilayer perceptron(MLP):

$$y = softmax(MLP(C^{(l)}(C^{(l-1)}(\dots C^{(0)}(G^{(0)}))))) \tag{8}$$

where y is the disease state predicted by the model for graph, and C stands for the cluster layer.

Clinical scores are continuous data, and the value of score represents the severity of related functions. Therefore, we used supervised regression task to extract clinical score information.

For the same classification task, the output of regression is as follows:

$$S = Relu(MLP(C^{(l)}(C^{(l-1)}(\dots C^{(0)}(G^{(0)}))))) \tag{9}$$

where S is the clinical scores vector predicted by the model.

3.3.3. Loss function

The training of the model mainly involves three tasks: graph classification, nodes clustering and clinical scores regression.

Here, we used the cross entropy loss to evaluate the classification task:

$$L_s = -\frac{1}{M}\sum_{m=1}^{M}(y'_m \log(y_m) + (1 - y'_m)\log(1 - y_m)) \tag{10}$$

where M is the subject capacity, y_m is the real label, and y'_m is the model prediction label.

We take Q as the distribution assigned to each subject. Inspired by the deep clustering [25], we encouraged models to perform high-confidence clustering to learn transcendental feature embeddings in potential spaces. The target distribution $P_{ij}^{(l)}$ is calculated as follows:

$$P_{ij}^{(l)} = \frac{\left(Q_{ij}^{(l)}\right)^2 / \sum_i Q_{ij}^{(l)}}{\sum_{j'}\left(Q_{ij'}^{(l)}\right)^2 / \sum_i Q_{ij'}^{(l)}} \tag{11}$$

Kullback–Leibler (KL) divergence can update the model in a gentler way, reducing the impact of severe noise on feature embeddings. Therefore, we used it to define the unsupervised loss which is as follows:

$$L_u = \sum_{l=1}^{L}\sum_{n=1}^{N}\sum_{k=1}^{K} P_{ij}^{(l)} log(\frac{P_{ij}^{(l)}}{Q_{ij}^{(l)}}) \tag{12}$$

where L is the number of GNNs layers, N is the number of nodes, and K is the number of cluster centroids.

The aim of the regression mission is to predict several clinical scores that primarily affect PD, as previously described. It encourages the model to pay attention to the features associated with disease states. As PD is a disease with high specificity, intra-class clinical score differences may not be ignored, so the root mean square error (RMSE) is used here to reduce the impact of outliers on model updates. The regression loss is defined as follows:

$$L_r = \frac{1}{2M}\sum_{i=0}^{M}(S_i - S'_i)^2 \tag{13}$$

where S_i is the predicted clinical scores, and S'_i is the ground truth.

In summary, the overall objective loss of training is:

$$L = L_s + (1 - \lambda_1)L_u + \lambda_2 L_r \tag{14}$$

where λ_1, λ_2 are the corresponding training parameters for the unsupervised loss and regression loss of the model, respectively.

Considering that the training will be unstable if we update the centroid with the same frequency as the network parameters. We set the parameter λ_1, and for each epoch we only consider cluster loss at the end of training phase. Dynamic adjustment combined with the two-stage training strategy: that is, the model centroid number is set to a fixed size mode:

1. When the current cluster centroid number K is greater than 1, $\lambda_1 = 1$;
2. When the current cluster centroid number is $K = 1$, $\lambda_1 = 0$.

4. Experiment

4.1. Experimental setting

We evaluated our MNC-Nets on the PD public dataset to answer the following three questions:

1. Is node clustering-based graph learning suitable for brain network classification tasks?
2. Does the addition of clinical scores benefit the learning of disease brain networks?
3. Can the model give interpretive results to aid clinical diagnosis?

4.1.1. Dataset

We tested the proposed method on the Parkinson's Disease Progression Markers Initiative (PPMI) dataset [26]. The PPMI is a comprehensive international study dedicated to identifying biomarkers associated with PD progression to deepen our understanding of the causes and mechanisms, and to find key tools to improve the success of PD treatment trials. We focused on using dMRI to discriminate individuals with prodromal Parkinson's disease from HC.

Table 1
The detail demographic data of subjects used in this paper.

Group	HC(97)	PPD(97)
Male/Female	56/41	38/59
Age[a]	60.91 + 9.44	65.33 + 7.68
MOCA[a]	27.75 + 1.71	27.57 + 1.96
MDS UPDRS1[a]	0.52 + 1.34	1.32 + 1.83
MDS-UPDRS2[a]	0.15 + 0.50	2.10 + 3.88
MDS-UPDRS3[a]	0.71 + 1.62	3.36 + 6.58

[a]mean ± standard deviation.

Table 2
The parameter settings of the model of MNC-Net.

Parameter name	Parameters
Optimizer	Adam
Learning rate	0.0002
Dropout rate	0.1
Batch size	5
Max training epoch	200
Graph embedding layers	2

We used 194 subjects with dMRI and sMRI from PPMI, including 97 HC and 97 patients with prodromal PD (PPD). Here, PPD refers to the stage with symptoms and signs, but not enough to define the type of disease. In addition, we extracted four clinical scores of each subject which evaluate their motor symptoms or non-motor symptoms. Specifically, we use the Montreal cognitive assessment (MOCA) [27] as cognitive scores and the Movement Disorders Society modified Unified Parkinson's Disease Rating Scale Part I Patient Questionnaire (MDS-UPDRS1) score as non-motor symptom score, MDS-UPDRS Part II Patient Questionnaire (MDS-UPDRS2) as daily exercise score, MDS-UPDRS Part II (MDS-UPDRS3) as the composite movement disorder score [28]. The details of population information such as gender and age used in the experiment are shown in Table 1.

4.1.2. Evaluation metrics

We applied a 10-fold cross-validation strategy to test all methods. Specifically, we random divided the subjects into 10 folds, 9 for training and 1 for testing. Repeated this for 10 times to get the average performance as final result. There are four matrices used to evaluate the classification performance: accuracy (ACC), F-scores (F1), precision (Prec), recall(Rec).

4.1.3. Implementation details

In our experiment, we evaluated the effectiveness of MNC-Net on PPMI datasets. The parameter setting of our method is shown in Table 2. For node cluster learning, the number of cluster centers k and weight λ_2 are very important hyperparameters that affecting downstream tasks' performance. Here we set them to 10 and 2, respectively, further exploration will be explained in Section 4.2.2. Besides, all competing methods models are parameterized with their best settings from the original papers.

Regarding software versions, we install CUDA 9.2 and PyTorch 1.7.1 with Python 3.6.11 are used. Note that all the experiments are running a Linux Server with the Intel(R) Xeon(R) E3-1231 v3 @ 3.40 GHz CPU, and the GeForce GTX TITAN GM200 GPU. Our code is available at: https://gitee.com/yxfamy/mnc-net_master.git.

4.2. Results

To fully evaluate the proposed method, we compared it with the TML method and other DL methods to evaluate the classification performance.

For TML take the flat SC matrix as features, and the dimension is N_2, where N_2 is the number of ROIs. These methods include random forest (RF), SVM and MLP.

Table 3
Comparison of classification performance with other methods.

Methods	ACC (%)	F1 (%)	Prec (%)	Recall (%)
RF	62.80	49.00	62.80	40.10
SVM	68.61	68.71	68.60	69.60
MLP	69.85	69.40	69.80	71.80
CNNs	70.46	69.50	70.50	71.30
Resnet50	68.75	69.50	70.50	71.30
GCN	78.29	79.82	84.57	78.26
SAG	86.57	86.66	89.57	85.42
HGP	81.86	79.14	80.57	85.19
Proposed	**95.50**	**95.49**	**97.00**	**94.42**

In this paper, we mainly focused on SBNs analysis. Thus, we did not consider pixel-based methods, but only DL suitable for brain network analysis, including Resnet50 [29], GCN [30], self attention graph pooling(SAG) [31] and hierarchical graph pooling with structure (HGP) [32]. The features of Resnet50 are normalized SC, while the input of graph methods are the same as our method.

4.2.1. Classification performance

As shown in Table 3, we reported the disease classification performance of our method and other competitive methods, where the best results are in bold. We got the following three important observations.

First, compared with TML methods, DL have more advantages in brain graph classification tasks, and are significantly better than the former in four assessments. This may be due to the complexity of the brain network itself, which requires a larger number of parameters for analysis.

Second, all graph methods performed better than non-graph methods. This is because most of the existing non-graph methods can only extract the representation from the data itself, ignoring the important structural information in graph data, which limits the generalization ability of the model [33].

Third, compared with other graph pooling methods, the proposed method has better classification performance. This may be because our model takes advantage of the community nature of ROIs in brain networks to integrate information in subnetworks more effectively. In addition, the inclusion of clinical score information in training encouraged models to learn more robust disease features and improved model generalization.

4.2.2. The effectiveness of the hyperparameters

For the MNC-Net, two critical components are the structure learning and multi-task learning, which included two hyperparameters: the number of centroids k and the regression loss weight λ_2. In order to explore the influence of them on the performance, we used the train set and test set to tune k and λ_2 of model.

First, we investigated the effect of k choice with λ_2 fixed at 0. The results shown in Fig. 3, where we illustrated results of the five main groups of parameter in the test set. We noticed that the number of centroids of k had significant effect on the classification performance, which means that the clustering scale is an important factor affecting structure learning. Specifically, the smaller the k, the more concentrated the subgraphs learned by the model. On the contrary, the higher the k is, the more distinguishable the features are until they exceed a certain range.

Second, we fixed k to 10 and varied λ_2. Because the regression task affects the updating of model parameters through loss, and the regression loss is close to one tenth of the classification loss. So in order to make the method work, we set λ_2 to start from 1 and the span is 0.5. As shown in Fig. 4, in the range of 1.0 to 3.0, the accuracy is the highest at $\lambda_2 = 2.0$, which is better than the model in the condition ($K = 10, \lambda_2 = 0.0$). This may be because λ_2 can be used as regularization between the subject and ROIs to alleviate the overfitting problem of the

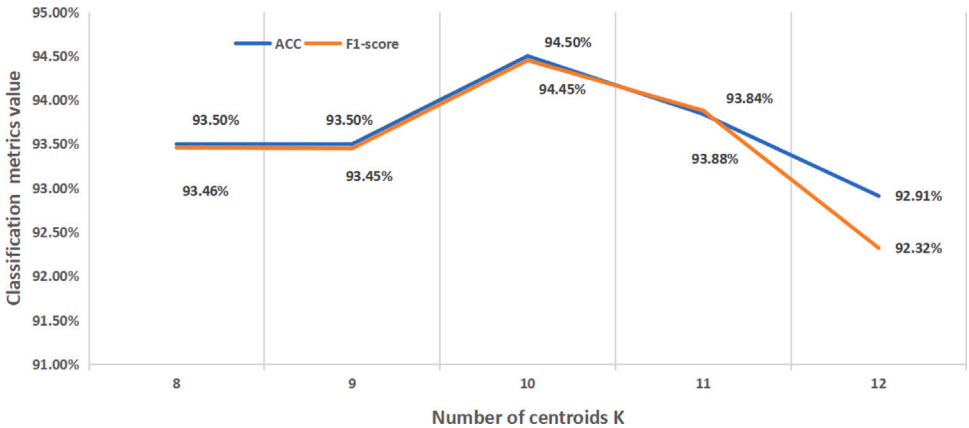


Fig. 3. Hyperparameter discussion(1): the classification performance for different number of K, when setting $\lambda_2 = 0$.

Fig. 4. Hyperparameter discussion(2): the classification performance for different number of λ_2, when setting $K = 10$.

model. Based on the tuning results for k and λ_2, we selected $K = 10$ and $\lambda_2 = 2.0$ as the best settings for the experiment.

The tuning results shown that the structure learning and the multi-task learning are effective for brain graph representation.

4.2.3. The effectiveness of proposed structure learning

To further illustrate the effectiveness of structure learning, we used t-distributed stochastic neighbor embedding (t-SNE) [34], which can project the multi-dimensional feature onto a two-dimensional space, to visualize the discriminability of representation. we extracted the output of the last classification layer as the features of each graph. The Fig. 5 shown the visualization results of Raw-graph, HGP, SAG and proposed MNC-Net on the PPMI dataset, where blue represents PPD, and red represents HC.

As shown, the SAG results are less satisfactory since the points with the same color are more dispersed. For HGP, the same type points is relatively compact, but the border between different class subjects distribution is still not so clear. Obviously, the visualization effect of our MNC-Net is much better, which represents the highest intra class similarity and forms a more compact cluster. The visualization results verify that the proposed method can learn better features, and prove the effectiveness of our structure learning.

4.2.4. The effectiveness of clinical scores regression

As an important indicator of PD diagnosis, clinical score can evaluate different functional states of the brain. Different scores may correspond to different brain regions. Therefore, we conducted relevant ablation experiments to discuss the effect of clinical scores on the proposed model. The results are shown in Table 4, and the best results are boldfaced.

Table 4
Effectiveness of different clinical scores on structural network learning. Total means all four scores participate in the training.

Scores	ACC (%)	F1 (%)	Prec (%)	Recall (%)
None	93.50	93.46	92.50	94.27
MOCA	93.50	93.51	95.00	92.18
UP1	94.00	93.94	94.50	93.77
UP2	94.00	94.00	96.00	92.36
UP3	93.00	92.94	92.50	93.59
Total	**95.50**	**95.49**	**97.00**	**94.42**

We noticed that when only one clinical score is considered, the model classification performance is lower than that of the original model. However, when all scores are taken into account, model performance improved significantly. This may be due to the heterogeneity of PPD. Brain network abnormalities tend not to focus on a single substructure, but distributed in different subnet. Therefore, considering only a single score may only identify specific types of PPD subjects, limiting the generalization performance of the model. Instead, we considered multifaceted clinical scores, which are consistent with the characteristics of PPD and more comprehensively guide the learning of SBNs.

4.3. Interpretability

One of the great advantages of GNNs is their built-in interpretation. In this paper, we attempted to improve interpretability by learning to

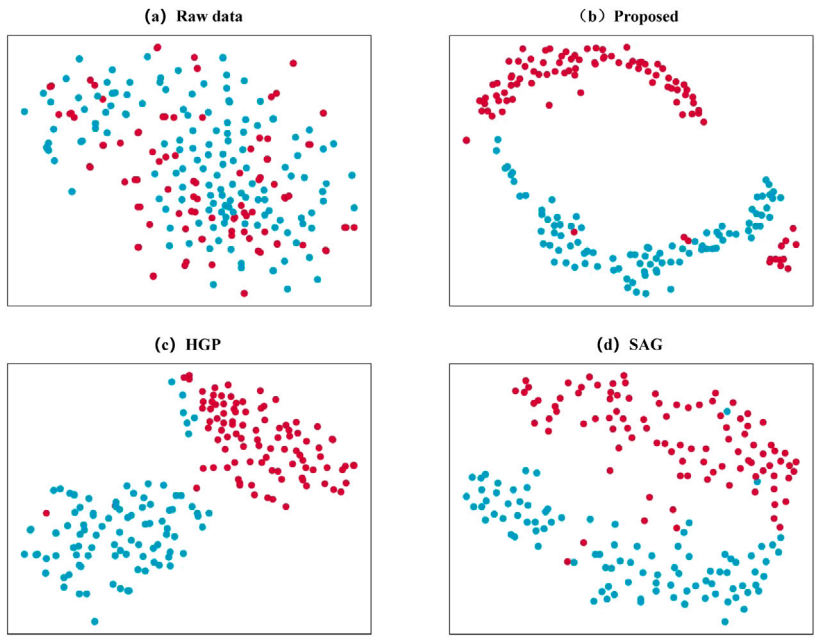


Fig. 5. T-SNE visualization on PPMI dataset. (a) Raw features; (b) learned by our proposed model; (c) learned by HGP; (d) learned by SAG.

identify key subnetwork representations through the community characteristics of nodes. We referred to the hypothesis suggested by [35] that meaningful explanations are more likely to come from a model with high classification accuracy. Intuitively, model interpretability is learning how a good teacher makes good decisions, not how bad decisions are made. Therefore, the highest classification accuracy model obtained in Section 4.2.1 are used for interpretability experiments.

4.3.1. Salient structural ROIs

In our work, in order to verify whether the model learned abnormal network patterns of the disease, we attempted to extract significant ROIs from them and validated against the existing literatures of relevant studies. Specifically, for each correctly classified subject in the test set, we took the top 10 nodes with the highest scores in the clustering layer as the set of significant ROIs for that subject. The scores of cluster layers of node v_i are:

$$Score_i = MAX\left(Q_{ij}^{(l)}\right), j \in [0, K] \tag{15}$$

where MAX is the maximum operation. Finally, the results of all subjects of the same kind were averaged, and the 10 ROIs with the highest frequency of occurrence were taken as the final disease-related salient structural ROIs.

As shown in Figs. 6–7, we visualized the top 10 brain regions of subject at different stages by BrainNetViewer [36]. We can see that the two subject groups share many regions, including the precentral gyrus cortex, supplementary motor areas(SMAs), insula-covered inferior frontal gyrus, orbitofrontal cognitive cortex, and insula, which correspond to early symptomatic cognition [37], emotion perception [38], and autonomic motor deficits in middle PD [39–41]. Moreover, these overlapping brain regions indicated that the network can capture disease-related abnormal brain patterns through clustering learning and effectively use them for brain network classification tasks.

These brain regions are further analyzed in the discussion section.

4.3.2. Relevance to clinical testing

To verify the significance of the extracted brain regions, we performed a more detailed statistical analysis using clinical scores. First, we merged the left and right parts of the above prominent ROIs to summarize the corresponding functions they perform. Second, we extracted clinical score subterms that evaluated the function of these ROIs and used paired t-tests (HC vs. PPD, significance level 0.95) to verify whether the model learned abnormal connections that distinguished PPD from HC. When the p-value is above 0.05, the difference between the two groups is statistically significant, meaning that the model acquired significant abnormal brain regions of PPD. Table 5 shown the statistical differences of HC and PPD data extracted from clinical scores. To facilitate display, we serialize the following ROI numbers: 1 (hippocampus), 2(Posterior cingulate gyrus(PGG), 3 (insula), 4 (precentral cortex), 5 (SMAs), 6 (middle frontal gyrus), 7 (superior occipital gyrus), 8 (medial superior frontal gyrus), 9(inferior frontal gyrus of insular part), and 10 (orbital frontal cognitive cortex).

Table 5
The clinical scores statistic related to salient ROIs.

ROIS	1,2,3	4,5,6,7	8,9,10
Functions	Cognition Emotion	Self-movement Trunk control	Speech-expression Speech-comprehension
Scores	UP1	NP2FREZ,NP2WALK NP2RISE NP2TRMR,NP2TURN	UP2SPCH UP3SPCH
P-value	<0.05	<0.05	<0.05

As shown in the figure, PPD and HC showed significant differences in the scores of cognition, emotion management, language expression and mild motor function. Symptoms corresponding to these dysfunctions often appear in the early stage of PD,when diagnosis and treatment are conducive to delaying the course of the disease.

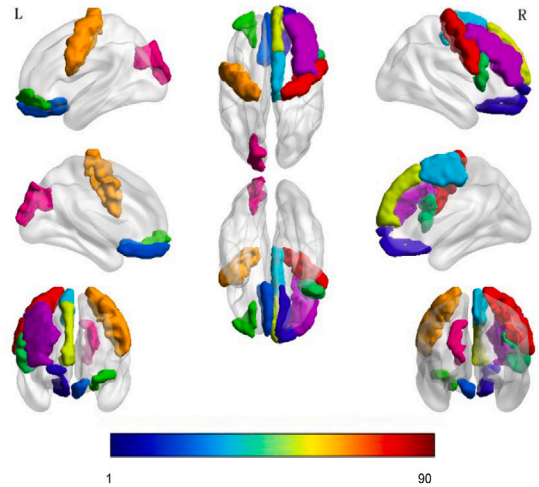
5. Discussion

5.1. The classification performance of model

In this paper, we proposed a multi-task brain network representation framework for early-stage PD classification. Our proposed framework includes (i) graph modeling of SBNs to transform complex brain image features to a learnable graph representation; (ii)replaced graph

Index	ROI Name	Score
2	Precentral gyrus-L	1.00
6	Superior frontal gyrus-orbital	0.96
8	Middle frontal gyrus	0.91
9	Middle frontal gyrus-orbital	0.78
27	Gyrus rectus	0.49
12	Inferior frontal gyrus-opercular	0.47
49	Superior occipital gyrus	0.39
20	Supplementary motor area-R	0.34
1	Precentral gyrus-R	0.30
24	Superior frontal gyrus-mideal	0.26

(a)

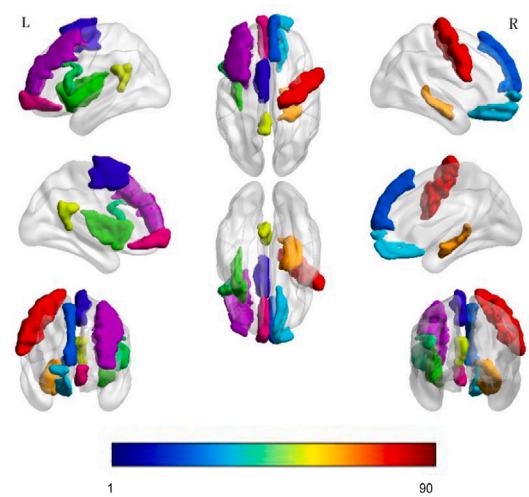


(b)

Fig. 6. Top 10 salient brain regions for PPD using proposed method. (a)Table containing ROI index, ROI name and Score (b)Visualization of salient brain regions.

Index	ROI Name	Score
19	Supplementary motor area	0.85
2	Precentral gyrus	0.74
6	Superior frontal gyrus-orbital	0.70
24	Superior frontal gyrus-mideal	0.59
7	Rolandic operculum	0.59
29	Insula	0.48
35	Posterior cingulate gyrus	0.44
38	Hippocampus	0.40
25	Superior frontal gyrus-medial orbital	0.37
11	Inferior frontal gyrus-L	0.30

(a)



(b)

Fig. 7. Top 10 salient brain regions for HC using proposed method. (a)Table containing ROI index, ROI name and Score (b)Visualization of salient brain regions.

pooling layers with graph aggregation and node clustering to reduce the effect of noise and enhance the representation of important connections. It shown good performance on PPMI dataset and achieved an average accuracy of 95.5%. In Section 4.2.1 we have demonstrated the relative performance of the proposed method.

To compare the model's classification performance at a larger scale, we compared it with related work based on SPECT. SPECT is the most effective imaging tool in the clinical diagnosis and treatment of PD, but it has the characteristics of high cost and invasive. In recent years, many researches have attempted to use non-invasive, inexpensive and more convenient detection tools to obtain results close to SPECT. Therefore, we used it as a unified criterion for evaluating the practicality of the proposed model. We compared the proposed method with related studies on the early classification of PD, as shown in Table 6. The results shown that our method achieved the best results except for the invasive method (SPECT). Furthermore, our method achieved accuracy close to invasive methods, illustrating the potential importance of DTI in understanding structural changes in the brain in PD. In the future, we can take further studies to use non-invasive imaging to assist clinicians in the diagnosis of early PD.

Table 6
The PD classification performance comparison of the proposed method and related work.

Study	Method	Modality	Dataset(%)	ACC(%)
Prashanth et al. [42]	SVM	SPECT	181 HC,493 PPD(PPMI)	97.29
Long, D. et al. [43]	LDA	T1	27 HC,19 PPD(Personal)	86.96
Amoroso et al. [44]	SVM	T1	169 HC,374 PPD(PPMI)	88.06
Liu et al. [45]	SVM	DTI	41 HC,64 PPD(PPMI)	76.32
Proposed	GNNs	DTI	97 HC,97 PPD(PPMI)	95.50

5.2. Interpretability

5.2.1. Salient structural ROIs

The other important purpose of this study is to find neuroimaging indicators that can distinguish patients with PD from HCs. Figs. 6–7 shown the salient ROIs such as the hippocampus, orbital frontal lobe, SMAs, PCG, etc. Previous studies have also demonstrated the validity of our extracted salient brain regions.

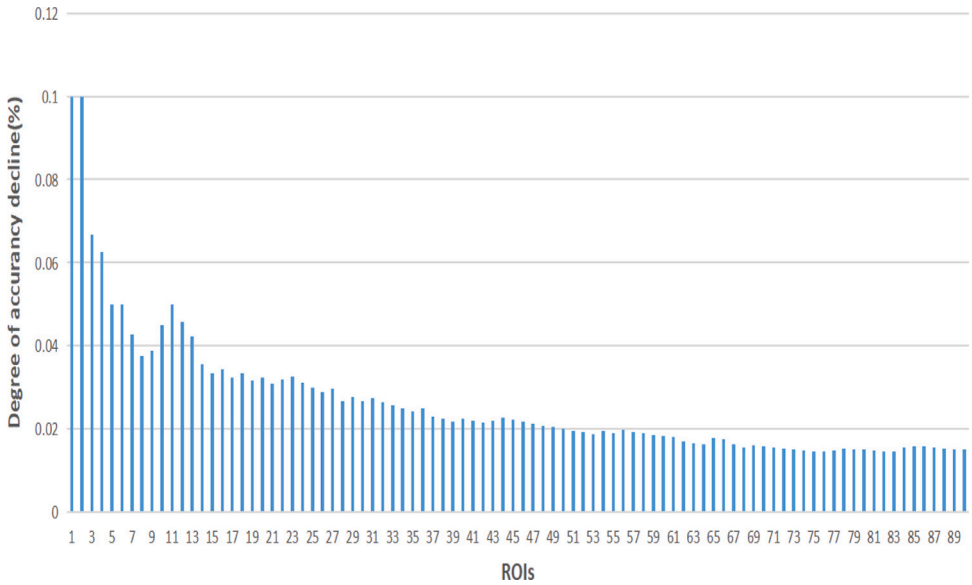


Fig. 8. Results of accuracy degradation of all ROIs based on mask method.

Györfi et al. [38] found that the hippocampus is responsible for long-term memory, and its longitudinal fiber bundle has different characteristics. Clinically, the hippocampus is involved in emotion regulation, stress, and anxiety management. Its synaptic plasticity damage with regional neural circuits, including the posterior cingulate gyrus and insula, may lead to cognitive and emotional disorders in early PD. Moreover, DTI studies [46] suggested that the orbital frontal lobe is associated with extensive white matter microstructural lesions of the limbic system in early PD with depression. These findings suggested that there may be abnormalities in the brain structure of memory, emotion management and cognition in early PD, which need to be paid attention to in diagnosis. In addition, the SMAs is part of the motor precortex located on the inner surface of the anterior central sulcus anterior cortex. Morris et al. [47] found that the SMAs is associated with spontaneous movement disorders such as tremors, and decreased SMAs activity (dopamine deficiency) is the cause of movement disorders in PD. PCG is the main motor cortex responsible for controlling autonomous movement. It contains a large number of large pyramidal cells, which are the center of motion and manage the random movement of the contralateral half of the body [39]. When it is injured, it may lead to contralateral hemiplegia Previous studies [40,41] found that the language organization, comprehension function, and spatial visual cognitive ability of PD patients decreased, and the relevant areas included insular lid medial superior, inferior frontal gyrus, and orbital frontal cognitive cortex. These indicated another pathological trend in early PD — mild motor symptoms. These symptoms may be considered normal aging, and observing SBNs connectivity may help distinguish between the two.

5.2.2. Exploration of significant regions based on mask

To further verify the significance of these ROIs, we used a masking based approach to explore the importance of each ROI to the classification model. The method based on occlusion comes from image detection. The core idea is to mask some pixels, and then observe the changes in the model's output. When extended to graph classification, we can verify whether the obtained model learns important features by masking the node features of the graph. Specifically, we hid all connections associated with one ROI in the samples in the test set. Then the trained model is used for testing, and the decrease in accuracy will be taken as the importance of ROI. We used brain templates with 90 ROIs in the experiment and needed to do more than 90 experiments. The results are as shown in Fig. 8.

The figure shown that the brain regions with the most significant impact on classification are brain regions 1 and 2, corresponding to the name of the precentral gyrus. For other brain regions with greater influence, we proposed significant ROIs overlap in Section 4.3.1. We put forward the significant ROI importance of the two groups as shown in Fig. 9–10.

As shown in the figure, the overall trend of ROIs importance ranking is not completely consistent with the score given by the model. This may be because the score considers not only the single node, but also the clustering characteristics of nodes. While the importance ranking mainly depends on the characteristics of a single node. For example, in Fig. 9, node 1 is ranked ninth, but its importance is higher than that of the preceding and posterior nodes. This may be because the templates treat the left and right parts of the same brain region as different regions(nodes 1 and 2 are the left and right brain parts of the precentral cortex), but they function similarly and therefore have equal importance. However, due to individual differences, the adjacent nodes of the two nodes are not entirely the same, so the model scores are different.

6. Conclusion

In this paper, we proposed a new MVC-Net framework for early diagnosis of PD. We modeled disease diagnosis as a graph classification problem to better incorporate its topological and SC properties. Based on the construction of SBNs in DTI, we defined the nodes and attributes of the subject graph. Inspired by the modularity of SBNs, we learned the abnormal pattern of the PPD brain network by node clustering and classified early PD based on high-level embeddings. In addition, we explored the relationship between brain structure, clinical scores, and subjects. By predicting the main clinical states of the subject, we encouraged the model to learn the effective features of the network and consider the specificity of the subject. We validated our method on the PPMI dataset. Our method shown superior accuracy for PPD classification and graph embedding compared with the TML, CNNs and GNNs methods. As shown in Table 3, the proposed method can effectively identify PPD from HC with an accuracy of 95.5%. By studying the clustering layer, we obtained the significant ROIs of PPD and HC group respectively. The interpretability of the proposed method is proved by interpretive experiments.

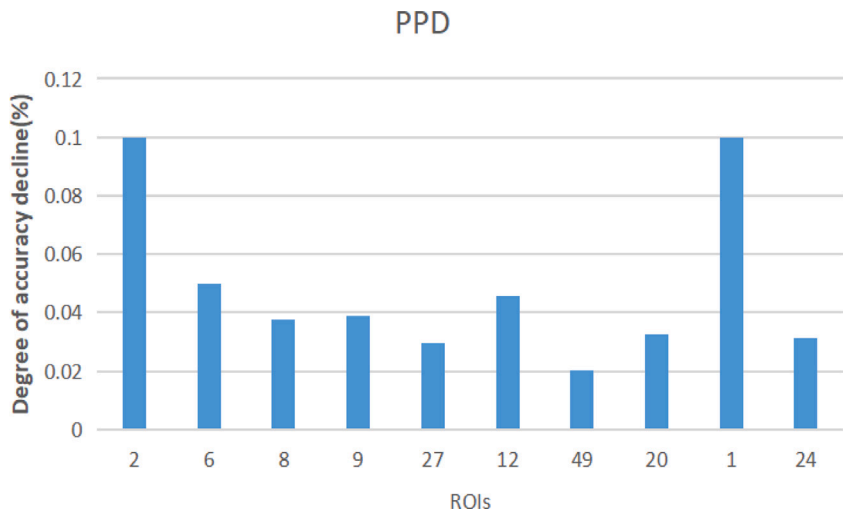


Fig. 9. Results of the degree of accuracy degradation corresponding to the top 10 significant ROIs of PPD.

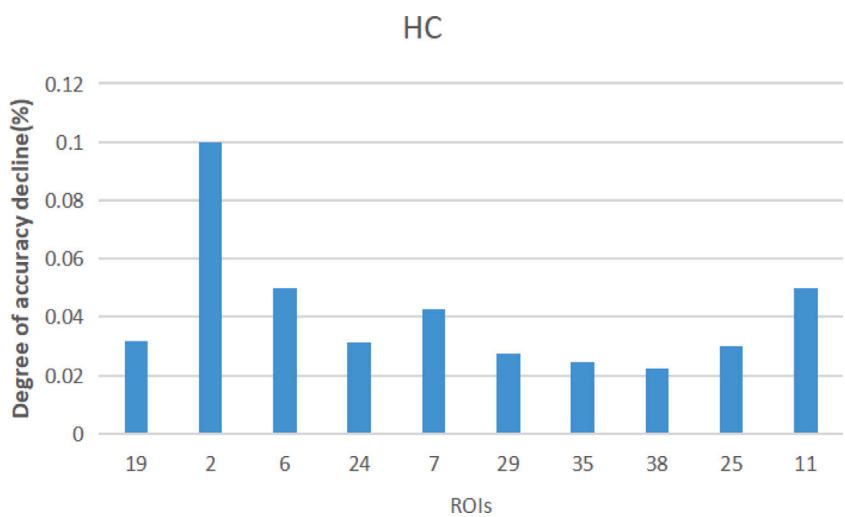


Fig. 10. Results of the degree of accuracy degradation corresponding to the top 10 significant ROIs of HC.

6.1. Limitation and future work

Although our method achieved promising performance on the PPMI dataset, there are still some limitations in our work. The pretreatment process in the experiment is merely an optional metric FA based on DTI, which is primarily isotropic specificity of brain tissue. However, from DTI, we can also obtain MA, L1, and other indicators, which may have PD-related information and have been used in some studies [48,49]. Therefore, our further work will be to use a more comprehensive feature to construct the multiview representation of ROI. End-to-end joint training with multiview graph node features is a challenge. Recently, some work has proposed an interesting idea to build multi-view features into heterogeneous graphs to improve model performance through higher-order neighbor aggregation [50] and recursive relationship fusion [51]. However, these works are aimed at node-level tasks and cannot be directly generalized to graph-level tasks. In future work, we will attempt to construct a dataset of brain heterogeneous graphs and analyze its impact on PD classification and interpretation at the graph level.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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